

M000 Week 0 — English Worksheet (Beginner)

Topic: Pixel Art — Counting Squares · **Course:** Builder Basics · **Level:** Beginner · **Time:** about 30 minutes

A block in Minecraft has three numbers: **X** (left ↔ right), **Y** (up ↑ down), and **Z** (depth). On a flat wall, **Z stays the same** — you only change **X** and **Y**.

 This is a warm-up to do **before Week 1**. It helps you count squares.

1 · Read the Wall

Open the **Pixel Art** world. A flat wall has red pixel art on it.

- **X** = the columns, left to right →
- **Y** = the rows, bottom to top ↑
- **Z** = the depth. It stays the **same**, so we ignore it here.

Every red shape has a **START** and an **END**. This worksheet shows you how to count to them.

Find the smallest red mark on the wall. Circle one — is it 1 square or 2 squares? 1 · 2

2 · Find START and END

Two steps find any square's spot:

Step ① — Count across (→) for X. 0, 1, 2, 3 ... Step ② — Count up (↑) for Y. 0, 1, 2, 3 ...

The red square below is .

Y \ X	0	1	2	3
2				
1				
0				

- **Count across:** 0, 1, **2** → X = 2.
- **Count up:** 0, **1** → Y = 1.
- **START = (2, 1). END = (2, 1).**
- It is **one** square, so START and END are the **same**. 

Blocks

```
fill [  red wool ]  
  from world X 2 Y 1 Z 0  
  to world X 2 Y 1 Z 0
```

Python

```
blocks.fill(RED_WOOL, world(2, 1, 0), world(2, 1, 0), FillOperation.Replace)
```

This square is 1 wide and 1 tall. Circle one — are START and END the same or different? same · different

3 · Find START and END of a Line 🧭

A line lying down has a **START** on the left and an **END** on the right.

In the grid: ■ **START** · ■ **END** · ■ **in between**.

Y \ X	0	1	2	3	4
2					
1					
0					

- **Count across:** red begins at X **1** and ends at X **3**.
- **Count up:** red is in row **1** only → Y = **1** for both.
- **START = (1, 1). END = (3, 1).**
- It is **1 tall**, so the **Y is the same** (1) for START and END.

Fill the blanks: **START** = (__ , __) **END** = (__ , __)

Circle the number that is the **SAME** for **START** and **END**: X · Y

4 · Find the Difference 🐉

Both lines look almost the same. One colors **one** square. One colors **four**.

```
# Line A
blocks.fill(RED_WOOL, world(2, 2, 0), world(2, 2, 0), FillOperation.Replace)
```

```
# Line B
blocks.fill(RED_WOOL, world(2, 2, 0), world(3, 3, 0), FillOperation.Replace)
```

Which line colors only **ONE** square — A or B? Why?

5 · Fill the Gap

You want a line lying down, **3** squares long. It starts at **X = 1** and sits on **Y = 2**. One number is missing.

Y \ X	0	1	2	3	4
2					
1					

```
blocks.fill(RED_WOOL, world(1, 2, 0), world(__, 2, 0), FillOperation.Replace)
```

Word bank: 2 · 3 · 4

Write the missing END number (count across to the ):

6 · Tell Me What You Built 📷

Now switch to your homework world. Color **one single square**, then make **one short line**. When you finish, come back here.

Send a photo or video, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My square had START at X ... , Y ...

For one square, my START and END were ...

Then I made a line. Its START was ... and its END was ...

7 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) and show your wall. Talk like you are teaching a friend. Try to use these words: **X, Y, START, END, same**.

1. Show the square you colored.
2. Count to its **START** and **END** out loud and say why they are the same.
3. Show your line and count to its **START** and **END**.

Write what you will say in your video. You can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.